

Analyzing Correlation between Patient Attributes and Heart Disease

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Introduction:

The leading cause of death within the United States is heart disease, accounting for 919,032 deaths in 2023. That is the equivalent of a third of all US deaths, or a death every 34 seconds.(CDC, 2024) Aggregating the cost of health care services, medicines, and lost productivity due to deaths brings the economic damage of heart disease to \$417.9 billion USD from 2020 - 2021. The devastating effects and costs of CVD(cardiovascular disease) can be seen from the aforementioned points alone, but another factor to consider is that since 1950, CVD death rates from CVD have declined 60%, showing how prevalent an issue heart disease has always been, and even with the decline, still continues to be. (Newsroom, 2024) The decline of CVD death rates is believed to be due to improved diagnosis and treatment options, clearly highlighting the importance of diagnosis and awareness of common factors. Therefore, our goal within this project will be to analyze the correlation between patient attributes and heart disease.

Research Question:

Exploring correlations between our patient attributes and determining if they are statistically significant. Specifically, analyzing which of our predictors tend to predict heart disease effectively, which attributes would be the most useful for diagnosis, and by extension, most effective for diagnosis and awareness.

Objectives:

1. Compile/Clean Heart Disease Dataset: Converting categorical attributes into factors, and changing certain values to be easier to work with.
2. Check Data Quality and Association: Visualize and describe our data, and see if there is any association between our response variable and our predictors.
3. Fit a Logistic Regression model: To identify which patient attributes(if any) can be seen as statistically significant in predicting a difference in Heart Disease Presence.
4. Interpret Results: Conclude with our findings based on our data, and potential uses for our findings.

Data:

Our dataset is taken from UC Irvine's Machine Learning repository, and consists of 270 instances of patient data, with 13 columns of patient attributes (our predictors), and lastly a column (our target) that identifies whether or not the patient had heart disease present. Our dataset consists of clean data with no missing values, meaning we will most likely use all of our columns.

Variable Name	Data Type	Description
age	Numeric - Continuous	age in years
sex	Char - Categorical (originally binary)	sex(F == 0, M == 1)
chest-pain	Char - Categorical	chest pain type
rest-bp	Numeric - Continuous	resting blood pressure
serum-chol	Numeric - Continuous	serum cholesterol
fasting-blood-sugar	Numeric - Binary	fasting blood sugar > 120 mg/dl
electrocardiographic	Char - Categorical	resting electrocardiographic results
max-heart-rate	Numeric - Continuous	maximum heart rate achieved
angina	Numeric - Binary	exercise induced anigna
oldpeak	Numeric - Continuous	oldpeak = ST depression induced by exercise relative to rest
slope	Char - Categorical(originally int)	the slope of the peak exercise ST segment
major-vessels	Numeric - Categorical(originally int)	number of major vessels (0-3) colored by fluoroscopy
thal	Char - Categorical	thalassemia: 3 = normal; 6 = fixed defect; 7 = reversable defect
heart-disease	Numeric - Binary(response)	present/not present

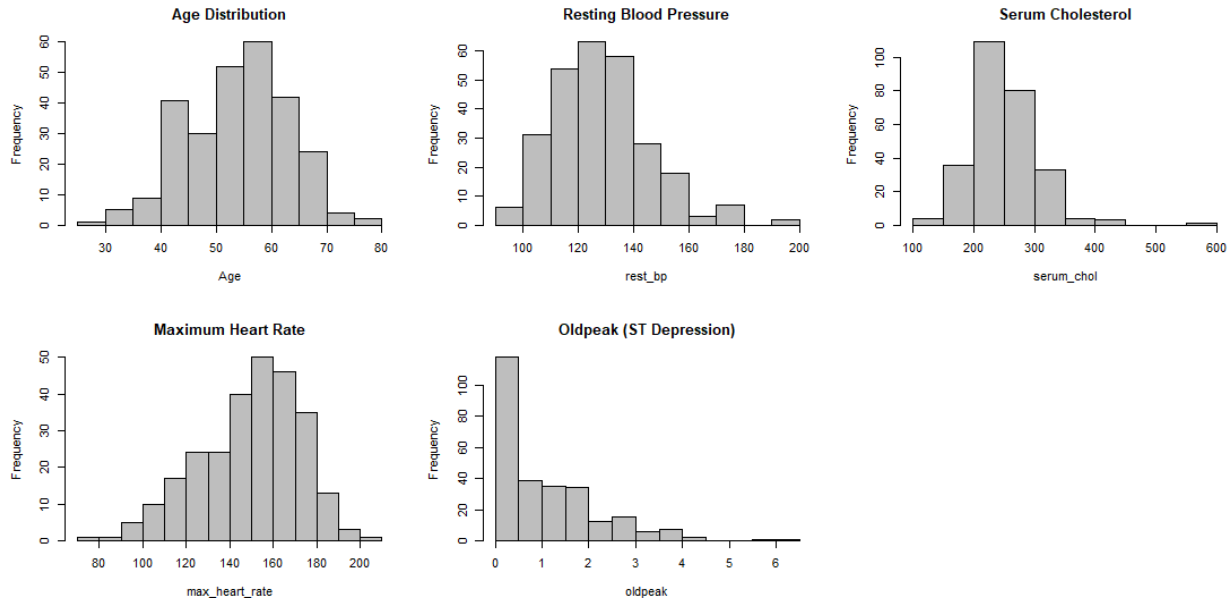
Data Quality Check:

Our set consisted of categorical, continuous, and binary variables. All categorical predictors were converted into the factor data type in R when conducting analysis, with missing factor categories clarified by a different UCI Heart Disease set.(UCI, 1988) Our response variable heart_disease is a binary variable (0 for not present, 1 for present), and following conversion to factors, this was our data summary:

Based on the summary, there doesn't seem to be any missing data, and categoricals have been converted to factors correctly. To gain a better understanding of our continuous columns, we started by checking the number of rows to determine our

population size($n == 270$). Due to this, we did not use the Shapiro-Wilk test, and instead we plotted histograms to check the distribution of our continuous predictors:

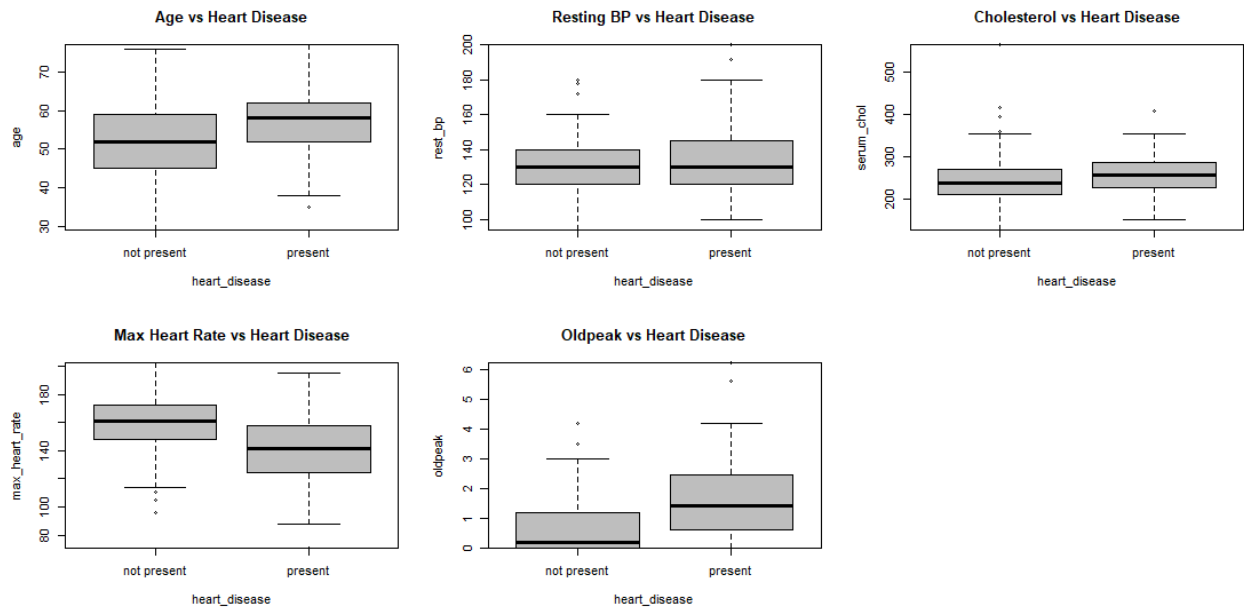
```
> nrow(data)
[1] 270
```



Evaluating the histograms, we can see that the distributions are all normally distributed or slightly skewed, other than Oldpeak, which is heavily skewed towards 0. This is a generally expected skew in healthy individuals, so it shouldn't affect our results.

Association(Response Vs. Predictors):

We continued by plotting our continuous predictors against our response variable using box plots, to determine if there was a potential association between the two by evaluating the box plots and seeing if the response variable showed different distributions.



Across the box plots, we noticed that every box plot other than Resting BP had noticeably different medians. This suggests that Age, Cholesterol, Max Heart Rate, and Oldpeak may be associated with heart disease status, and Resting BP seems to be similar across both groups.

We then tackled our categorical variables using a Chi-Square test. Our H_0 will be that our predictors are not significant under a 95% confidence interval, and our H_a will be that they are significant. Below are the results of the Chi-Square test in a table, including whether or not we rejected our null hypothesis under our 95% interval:

Variables	X^2	Deg of F	P-value	Null Hypo
sex	22.667	1	1.926e-06	Reject
fasting_blood_sugar	0.0091712	1	0.9237	Keep
angina	45.692	1	1.384e-11	Reject
chest_pain	68.588	3	8.561e-15	Reject
electrocardiographic	8.9795	2	0.01122	Reject
slope	40.37	2	1.713e-09	Reject

thal	74.569	2	< 2.2e-16	Reject
major_vessels	62.863	3	1.437e-13	Reject

For our Chi-Squared table, we found that all variables other than fasting_blood_sugar were associated with our response variable under our 95% confidence interval.

Logistic Regression Model:

We started off by using the glm() function to model heart disease probability off of our predictors. The following would be the equivalent equation and results:

$$\log((P(\text{Heart Disease}=\text{present}))/ (1-P(\text{Heart Disease}=\text{present}))) = \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{Sex}) + \beta_3(\text{Angina}) + \beta_4(\text{chest_pain}) + \dots$$

```
> summary(logit_hd)
```

Call:

```
glm(formula = heart_disease ~ age + sex + angina + chest_pain +
     max_heart_rate + oldpeak + slope + thal + major_vessels,
     family = binomial, data = data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-3.5337099	2.7575928	-1.281	0.200037	
age	0.0007923	0.0243123	0.033	0.974004	
sexmale	1.3820637	0.5320755	2.597	0.009391	**
anginayes	0.6120780	0.4453032	1.375	0.169280	
chest_painatypical angina	1.3131974	0.8650253	1.518	0.128988	
chest_painnon-anginal pain	0.3073734	0.7396752	0.416	0.677738	
chest_painasymptomatic	2.2968513	0.7392197	3.107	0.001889	**
max_heart_rate	-0.0121960	0.0110953	-1.099	0.271679	
oldpeak	0.5627814	0.2409242	2.336	0.019495	*
slopeat	1.0729304	0.4921022	2.180	0.029235	*
slopedownslopeing	0.2482683	0.9593449	0.259	0.795798	
thalfixed defect	-0.2464840	0.8116042	-0.304	0.761357	
thalreversible defect	1.4428153	0.4360032	3.309	0.000936	***
major_vessels1	1.9243842	0.5115379	3.762	0.000169	***
major_vessels2	2.5551779	0.7579729	3.371	0.000749	***
major_vessels3	2.0097452	0.8892477	2.260	0.023818	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 370.96 on 269 degrees of freedom
Residual deviance: 174.74 on 254 degrees of freedom
AIC: 206.74

Number of Fisher scoring iterations: 6

Results:

Based on our model's results, at the 5% significance level our statistically significant predictors included sex, chest_pain type (asymptomatic), oldpeak(ST depression), slope (flat), thal (reversible defect), and major_vessels(number of major vessels). We then converted our coefficients to odds-ratios for our significant predictors, and percentages.

When controlling for all other variables at constant, we found that within our dataset:

The odds for males of having heart disease present were 3.983 times higher than in females, or around a 298.3% increase.

The odds for patients with asymptomatic chest pain having heart disease present were 9.942 times higher than in patients with typical angina, or around a 894.2% increase.

The odds for patients per one unit increase in oldpeak(ST depression) of having heart disease present was 1.756 times higher, or around a 76% increase.

The odds for patients with a flat ST slope were 2.924 times higher than in patients with upsloping ST slopes, or around a 192% increase.

The odds for patients with a reversible thalassemia defect were 4.233 times higher than in patients with normal results, or around a 323.3% increase

The odds for patients with 1, 2, and 3 major vessels were respectively 6.581, 12.874, and 7.461 times higher than patients with 0 major vessels, or around respective increases of 585.1%, 1187.4%, and 646.1%.

Conclusion:

Our analysis found that most patient attributes in the dataset could be associated with the presence of heart disease. The chi-square tests flagged every categorical predictor except for fasting blood sugar, and the logistic regression model identified sex, asymptomatic chest pain, ST depression, a flat ST slope, reversible thalassemia defect, and number of major vessels as significant at the at the 95% confidence level.

The chest pain result carried the most weight. Patients presenting with asymptomatic chest pain could expect roughly ten times the odds of heart disease compared to patients with typical angina when holding other variables constant. That runs against the intuition that the most recognizable symptom is the best predictor, and it connects directly to the awareness problem: the American Heart Association reported that more than half of U.S. adults do not know heart disease is the leading cause of death (Newsroom, 2024). If people associate cardiac risk with predictors not most predictive of it, awareness campaigns built around typical symptoms may mislead the patients who need screening but think they are in the clear. Given that heart disease accounted for 919,032 U.S. deaths in 2023 (CDC, 2024), the cost of that gap to the nation and individuals is clearly substantial.

Several limitations bound the findings. The dataset contains 270 observations, which is a relatively small sample size for a model with this many predictors, and the odds ratios on major vessels suggest the estimates there are potentially unstable. The data is also cross sectional, so these are associations rather than causal effects. The underlying records date from 1988 (UCI, 1988), and diagnostic standards have also changed since. And these are patients already referred for cardiac testing rather than a general population, so the base rates here are likely inflated compared to what they would be in the community.

What the analysis does actually support is that a small number of measurable predictors carry most of the signal, which is useful for deciding which indicators deserve priority in both screening and in public awareness efforts.

References:

<https://www.cdc.gov/heart-disease/data-research/facts-stats/index.html>

(CDC, 2024)

<https://newsroom.heart.org/news/more-than-half-of-u-s-adults-dont-know-heart-disease-is-leading-cause-of-death-despite-100-year-reign>

(Newsroom, 2024)

<https://archive.ics.uci.edu/dataset/145/statlog+heart>

(UCI, 1988)